

Supplementary Material — The Changing Landscape of Medical Vision-Language Models: A Quantitative Survey of 1,217 arXiv Papers, 2024–2026

Repository (code, data, validation files): <https://github.com/fofifafo/medical-vlm-landscape-2026>

S1. arXiv queries (22). Each query was issued to the arXiv API over submissions dated 2024-01-01 to 2026-08-31; abs: restricts to the abstract field. Results were de-duplicated by arXiv identifier (latest version). Query 1 was split by year to stay below the API pagination limit.

Q1. abs:"vision-language model"

Q2. abs:"multimodal large language model"

Q3. abs:"visual instruction tuning"

Q4. abs:"efficient MLLM"

Q5. abs:"efficient vision-language"

Q6. abs:"token pruning" AND abs:multimodal

Q7. abs:"visual token" AND abs:compression

Q8. abs:"on-device" AND abs:"vision language"

Q9. abs:"mobile" AND abs:"vision-language model"

Q10. abs:"lightweight" AND abs:MLLM

Q11. abs:"KV cache" AND abs:multimodal

Q12. abs:quantization AND abs:MLLM

Q13. abs:"any resolution" AND abs:multimodal

Q14. abs:"high-resolution" AND abs:MLLM

Q15. abs:"document understanding" AND abs:multimodal

Q16. abs:"video" AND abs:MLLM

Q17. abs:"visual grounding" AND abs:"large language model"

Q18. abs:"multimodal hallucination"

Q19. abs:"multimodal" AND abs:"chain-of-thought"

Q20. abs:"retrieval-augmented" AND abs:multimodal

Q21. abs:"vision-language-action"

Q22. abs:"open-vocabulary" AND abs:segmentation

S2. VLM-relevance screen. A record was VLM-relevant when the title or abstract matched any of: multimodal large language model|MLLM|multi-modal LLM|multimodal LLM | vision[-]language model|VLM\b | visual instruction tuning (case-insensitive). 15,394 of 18,142 records.

S3. Medical filter. A record was medical when the following pattern matched the title, or matched the abstract at least twice (case-insensitive). 1,217 records.

medical|clinical|radiolog|patholog|patient|biomedical|chest x-?ray\b|CXR\b|CT scan\b|MRI\b|bEHR\b|hospital|healthcare|health care|physician|nursing|surgery|surgical|oncolog|dermatolog|ophthalmolog|endoscop|histolog|whole[-]slide|ultrasound|fundus\b|ECG\b

S4. Taxonomy patterns (title + abstract, case-insensitive): 39 tags in four axes. Table I shows 24 of them; the others were computed and are included in data/medical_vlm_corpus_1217.csv.

A. Clinical domain / modality

방사선 — X-ray/CXR: chest x-?ray\bCXr\b\radiograph|x-?ray

방사선 — CT/MRI (3D): \bCT\b\bMRI\b\volumetric|3D medical|computed tomography|magnetic resonance

병리 — 조직/WSI: pathology|histology|whole[-]slide\bWSI\b|biopsy

안과 — 안저/OCT: ophthalm|fundus|retina\bOCT\b

피부과: dermatolog|skin lesion|skin disease

내시경·수술 영상: endoscop|surgical (video|scene)|laparoscop|surgery video|colonoscop

초음파: ultrasound|sonograph|echocardio

생체신호 (ECG/EEG): \bECG\b\bEKG\b\bEEG\b|electrocardio

치과·구강: dental|oral (cavity|health)|tooth|teeth

전자의무기록·문서: \bEHR\b|electronic health record|clinical note|discharge summary|medical record

일반/다중 모달리티: multi[-]?modal medical|across modalities|diverse medical imaging|multiple imaging modalit

B. Task

판독문 생성: report generation|radiology report|generate (a |the)?report|report writing

의료 VQA: \bVQA\b|visual question answering|question[-]answering

진단·분류: diagnos|classif|disease (detection|prediction|identification)

병변 탐지·그라운딩: lesion detection|grounding|localiz|bounding box|abnormality detection

분할: segment

검색·RAG: retriev\bRAG\b|retrieval[-]augmented

임상 의사결정·추론: clinical (decision|reasoning)|differential diagnosis|treatment (plan|recommend)|triage

의학 교육·시험: medical (exam|licens|education)|USMLE|board exam|medical student

환자 소통·상담: patient (communication|education|conversation)|health assistant|chatbot|consultation

수술 보조: surgical (assist|guidance|phase|skill)|intraoperative

C. Method

범용 VLM 파인튜닝 (LLaVA 계열): LLaVA|fine[-]tun\w* (a |the)?(general|open|pretrained)|adapt\w* (a |the)?general

도메인 사전학습 (CLIP 계): BiomedCLIP|MedCLIP|PubMedCLIP|contrastive|CLIP

명령어 튜닝·데이터 구축: instruction[-]tun|instruction data|instruction[-]following dataset|curated dataset|we (build|construct|introduce) (a|an)?(new |large[-]scale)?dataset

검색 증강: retrieval[-]augmented|bRAG\b|knowledge (base|graph) (augment|enhanc|inject)

에이전트·도구 사용: \bagent|tool[-]use|tool[-]calling|multi[-]agent

추론 기법 (CoT/RL/GRPO/DPO): chain[-]of[-]thought|reinforcement learning\bRL\b\bGRPO\b\bDPO\b\bRLHF\b|R1[-]style|reasoning (model|trace|chain)

추론 언급 (bare, 어휘): reasoning

경량화·효율: efficien|lightweight|quantiz|distill\bLoRA\b|parameter[-]efficient|on[-]device|token (prun|compress)

연합·프라이버시: federated|privacy[-]preserving|differential privacy|de[-]identif

D. Stated concern

환각 (명시적): hallucinat

신뢰성 어휘 (넓게): hallucinat|factual|faithful|reliab

벤치마크·평가: benchmark|evaluation (framework|protocol|suite)|we evaluate|systematic evaluation

안전·오용: safety|jailbreak|adversarial|misuse|harm

공정성·편향: fairness|demographic|disparit|equit\b\bias(ed)? (against|toward|across)|algorithmic bias|racial|gender bias

설명가능성: explainab|interpretab|attention map|saliency

데이터 부족·희소: data[-]scarc|low[-]resource|few[-]shot|limited (annotat|label|data)

임상 검증·실사용: clinical (validation|trial|deployment|practice)|real[-]world|prospective|reader study|radiologist(s)? (evaluat|rat|prefer)

규제·윤리: \bregulat(ory|ion|ions|ors?)\b\bFDA\b\bethic|accountab\b|bliabilit

S5. Strict definitions (Table I rows 'Grounding contribution (strict)' and the clinical-validation count of Section VI).

grounding_contribution: (propose|present|introduce)[^.]{0,200}(grounding|localiz|bounding box|lesion detection|abnormality detection)

clinical_validation_strict: reader study|prospective|clinical trial|(radiologist|clinician|physician)s? (evaluat|assess)|evaluat\w* by (w+)?(radiologist|clinician|physician)

S6. Efficiency-oriented set (144). Papers carrying at least one of the twelve keyword tags efficient-*vlm*, lightweight-*vlm*, small-*vlm*, on-device, mobile-*vlm*, low-power, token-pruning, token-merging, kv-cache, quantization, knowledge-distillation, peft-lora, assigned by the controlled vocabulary in *code/arxiv_scrape.py*.

S7. Judging model. Claude Opus 5 (Anthropic), interactive session, September 2026; one pass per abstract (classification validation) or per full text (efficiency audit) with the written criteria as the instruction; outputs stored verbatim in validation/ and not re-generated.

S8. Classification-validation criteria (12 tags; validation/class_tags_guide.md).

Classification-validation criteria (12 tags)

A tag is credited only when the paper actually performs the activity; the loose regex used by the rule-based classifier is shown for comparison. Judged on title + abstract.

Tag | Criterion (credit only if the paper does this) | Rule regex (case-insensitive)

chest_xray | the paper works on chest X-rays (CXR / radiographs) | chest x-?ray\bCXR\b|radiograph|x-?ray

ct_mri_3d | the paper works on CT or MRI volumes | \bCT\b\bMRI\b|volumetric|3D medical|computed tomography|magnetic resonance

pathology_wsi | the paper works on pathology slides / tissue images | patholog|histolog|whole[-]slide\bWSI\b|biopsy

endoscopy_surgical | the paper works on endoscopic or surgical video/images | endoscop|surgical (video|scene)|laparoscop|surgery video|colonoscop

report_generation | report generation is a task of the paper | report generation|radiology report|generate (a |the)?report|report writing

grounding_localization | lesion localization / grounding is an actual task or contribution (the word alone does not count) | lesion detection|grounding|localiz|bounding box|abnormality detection

clinical_decision | clinical decision support or differential diagnosis is a task of the paper | clinical (decision|reasoning)|differential diagnosis|treatment (plan|recommend)|triage

reasoning_technique | a concrete reasoning technique is used: CoT, reinforcement learning, GRPO/DPO | chain[-]of[-]thought|reinforcement learning\bRL\b\bGRPO\b\bDPO\b\bRLHF\b|R1[-]style|reasoning (model|trace|chain)

agents_tool_use | an agent design or tool calling is used | \bagent|tool[-]use|tool[-]calling|multi[-]agent

efficiency_technique | an efficiency technique (compression, quantization, LoRA, ...) is actually applied (the adjective 'efficient' alone does not count) | efficien|lightweight|quantiz|distill\bLoRA\b|parameter[-]efficient|on[-]device|token (prun|compress)

hallucination_explicit | hallucination is addressed explicitly | hallucinat

clinical_validation | the paper reports a clinician evaluation, prospective validation, or real-world deployment (statements of intent do not count) | clinical (validation|trial|deployment|practice)|real[-]world|prospective|reader study|radiologist(s)? (evaluat|rat|prefer)

Key distinction: the presence of a word is not the same as doing the thing. 'grounded reasoning' without any localization → grounding X; 'efficient' without a compression technique → efficiency X; 'needed for real-world deployment' without an evaluation → clinical_validation X. A paper may carry several tags.

S9. Sampling. Classification validation: 150 papers, 50 per year, seed 20260915. Corpus-inclusion validation: 100 included papers (33/33/34 by year) plus 50 boundary exclusions (VLM-relevant records with exactly one clinical-term match in the abstract), seed 20260918; judged by the first author, disagreements with the rule adjudicated against the written criterion. Efficiency-audit verification: 30 papers stratified 15 with / 15 without a reported quantity, seed 20260914; all 180 judgments checked by the first author against the stored evidence sentences.

S10. Efficiency-audit rubric (validation/RUBRIC.md).

Full-text efficiency audit rubric (144 efficiency-oriented papers, 2026-09-14)

Purpose: extend Table II from the abstract level to the full-text level. For each paper and each quantity, judge only whether the body (tables, figures, sentences) reports a value that **the authors measured for their own system**.

Quantities

Code | Quantity | Counts as Y | Does not count

LAT | Latency / time | inference or training time of the authors' own model (ms or s per image, speed-up factors from their own experiments) | values cited from other papers; adjectives such as "real-time"; number of training epochs

HW | Target hardware | GPU/CPU/edge device used in the authors' experiments | devices mentioned in related work

FLOP | Compute | FLOPs or MACs of the authors' own model | the concept only

MEM | Memory | GPU memory / VRAM / footprint of the authors' own model | dataset size; model file size (bit-width alone counts as MEM only if a footprint is given)

ENG | Energy / power | measured W, J, mWh, power or carbon of the authors' own system | "energy-efficient" as an adjective; battery mentioned only

THR | Throughput | samples/s, QPS, fps of the authors' own model | —

PAR | (auxiliary) Parameters | total / active / trainable parameter counts | —

Values

- Y the body reports the authors' own measurement (supporting sentence required)
- A reported in the abstract only, nothing further in the body (same as the abstract-level audit)
- N not reported
- ? unreadable (PDF conversion failure, number only inside a figure) → resolved to Y/N after checking the page

Section rules

- Ignore everything after References. Sentences in Related Work / Background describe other papers and do ****not**** count, unless the subject is "we"/"our" and the authors' own model is named.
- Numbers in tables are assigned by caption and column name (e.g., a "Time (s)" column → LAT).

Outputs

- audit.csv: id, title, LAT, HW, FLOP, MEM, ENG, THR, PAR, any_Y, note (plus LAT_infer, the inference-side split)
- audit.jsonl: one line per paper with the supporting sentence for each Y
- Verification sample: 30 papers stratified by any_Y (15 with / 15 without), all 180 judgments checked by the first author against the stored sentences (validation_human.json, validation_kappa.json)

S11. Statistics. Fisher's exact test (two-sided) on 2024 vs 2026 counts per tag; Bonferroni $\alpha = 0.05/24$ over the Table I rows (all 41 tests, $\alpha = 0.05/41$, leave the same six rows significant); Cochran–Armitage trend test over 2024–2026 with scores 0/1/2; sensitivity analysis restricting 2026 to January–June. Implementation: `code/strict_tags.py`, `code/trend_tests.py`.

S12. Table II partition of the 144 efficiency-oriented papers: 28 report none of the six quantities; 65 name hardware and no other quantity; 11 report training time as their only quantity; 40 report at least one inference-relevant quantity (inference latency, FLOPs/MACs, memory, throughput). $28 + 65 + 11 + 40 = 144$. Of the 42 papers with any latency figure, 27 report inference latency and 15 training time only; 'two or more quantities' (7 abstracts, 25 full texts) counts quantities other than hardware.